

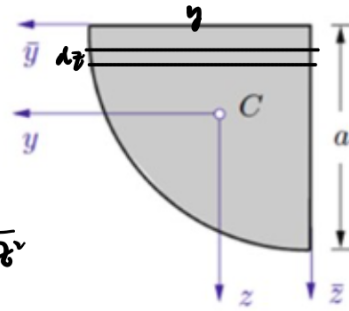
Homework 4.7 Determine the moments of area for the quarter-circular area:

- $S_{\bar{y}}, S_{\bar{z}}$, and
- $I_{\bar{y}}, I_{\bar{z}}, I_{\bar{y}\bar{z}}$, and
- I_y, I_z, I_{yz} about the centroidal axes y, z ;
- find the principal axes.

$$z^2 + y^2 = a^2$$

$$y = r \cos \theta \quad a = \sqrt{y^2 + z^2}$$

$$z = r \sin \theta$$



$$a) \int \bar{y} = \int z dA \quad \int \bar{z} = \int y dA \quad \text{Area of quarter circle} = \frac{1}{4} \times \pi a^2 = \frac{\pi a^2}{4}$$

$$\int \bar{y} = \int z dA = A z_c \quad \int \bar{z} = \int y dA = A y_c$$

$$\begin{aligned} \bar{z}_c &= \frac{1}{A} \int \bar{z} dA = \frac{4}{\pi a^2} \int \bar{z} dA = \frac{4}{\pi a^2} \int_0^{\frac{\pi}{2}} \int_0^a r \sin \theta \cdot r dr d\theta \\ &= \frac{4}{\pi a^2} \int_0^{\frac{\pi}{2}} \left. \frac{a^3}{3} \sin \theta \right|_0^a d\theta \\ &= \frac{4}{\pi a^2} \int_0^{\frac{\pi}{2}} \frac{a^3}{3} \sin \theta d\theta \\ &= \frac{4}{\pi a^2} \left(-\frac{a^3}{3} \cos \theta \right) \Big|_0^{\frac{\pi}{2}} \\ &= \frac{4}{\pi a^2} \left(-\frac{a^3}{3} \right) \\ &= \frac{4a}{3\pi} \end{aligned}$$

$$\bar{y}_c = \bar{z}_c = \frac{4a}{3\pi}$$

$$\int \bar{y} = \frac{\pi a^2}{4} \times \frac{4a}{3\pi} = \frac{a^3}{3} = \int \bar{z} \quad \text{because symmetry across } y=z \text{ line.}$$

$$\begin{aligned} b) I_{\bar{y}} &= \int \bar{z}^2 dA = \int_0^{\frac{\pi}{2}} \int_0^a r^2 \sin^2 \theta \cdot r dr d\theta = \int_0^{\frac{\pi}{2}} \int_0^a r^3 \sin^2 \theta dr d\theta \\ &= \int_0^{\frac{\pi}{2}} \frac{a^4}{4} \sin^2 \theta d\theta = \int_0^{\frac{\pi}{2}} \frac{a^4}{4} \left(\frac{1}{2} - \frac{1}{2} \cos 2\theta \right) d\theta \\ &= \frac{a^4}{4} \left[\frac{1}{2} \theta - \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{2}} \\ &= \frac{a^4}{4} \left(\frac{\pi}{4} - 0 - 0 + 0 \right) \\ &= \frac{\pi a^4}{16} \end{aligned}$$

$$I_{\bar{z}} = I_{\bar{y}} \quad \text{because of symmetry across } y=z \text{ line}$$

$$\begin{aligned} I_{\bar{y}\bar{z}} &= - \int \bar{y} \bar{z} dA = - \int_0^{\frac{\pi}{2}} \int_0^a r^2 \sin \theta \cos \theta \cdot r dr d\theta = - \int_0^{\frac{\pi}{2}} \frac{a^4}{4} \sin \theta \cos \theta d\theta \\ &= - \frac{a^4}{4} \left(\frac{1}{2} \right) \int_0^{\frac{\pi}{2}} \sin 2\theta d\theta = - \frac{a^4}{8} \int_0^{\frac{\pi}{2}} \sin 2\theta d\theta \\ &= - \frac{a^4}{8} \left(-\cos 2\theta \right) \Big|_0^{\frac{\pi}{2}} = - \frac{a^4}{8} \end{aligned}$$

$$c) \bar{I}_y = I_y + \bar{z}_c^2 A$$

$$I_y = \bar{I}_y - \bar{z}_c^2 A = \frac{\pi a^4}{16} - \left(\frac{4a}{3\pi}\right)^2 \left(\frac{\pi a^2}{4}\right) = \frac{\pi a^4}{16} - \frac{4a^4}{9} = 0.0549 a^4$$

$I_z = I_y$ due to symmetry

$$\bar{I}_{yz} = I_{yz} - \bar{y}_c \bar{z}_c A$$

$$I_{yz} = \bar{I}_{yz} + \bar{y}_c \bar{z}_c A = -\frac{a^4}{8} + \frac{4a^2}{9\pi} \left(\frac{\pi a^2}{4}\right) = 0.0165 a^4$$

$$d) I_{1,2} = \frac{I_y + I_z}{2} \pm \sqrt{\left(\frac{I_y - I_z}{2}\right)^2 + I_{yz}^2}$$

$$= 0.0549 a^4 \pm \sqrt{0 + (0.0165 a^4)^2}$$

$$= 0.0549 a^4 \pm 0.0165 a^4$$

$$I_1 = 0.0714 a^4, \quad I_2 = 0.0384 a^4$$

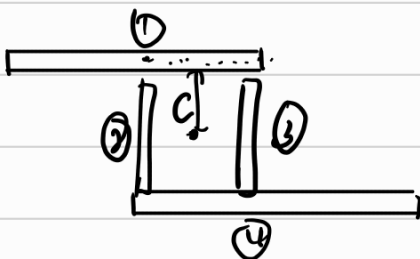
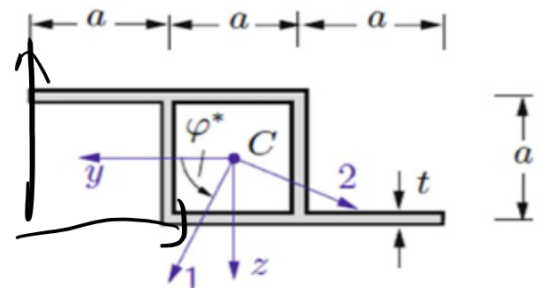
$$\tan 2\varphi^* = \frac{2I_{yz}}{I_y - I_z} = \frac{2I_{yz}}{0}$$

$$2\varphi^* = 90^\circ$$

$$\varphi_1^* = 45^\circ, \quad \varphi_2^* = 135^\circ$$

Homework 4.8 Determine the moments of inertia for the thin-walled cross section ($t \ll a$) about its centroidal axes and find the principal axes.

$$C\left(\frac{a}{2}, \frac{a}{2}\right)$$



$$I_y = I_{y1} + I_{y2} + I_{y3} + I_{y4} \quad I_{y1} = I_{y4} \quad I_{y2} = I_{y3}$$

$$= \left[\frac{1}{2}(2a)(t^3) + \left(\frac{a}{2}\right)^2(2at) \right] \times 2 + \frac{1}{12}a^3t \times 2$$

$$= 2\left(\frac{a}{2}\right)^2(2at) + \frac{1}{6}a^3t = a^3t + \frac{1}{6}a^3t = \frac{7}{6}a^3t$$

(neglect higher order terms of t as $t \ll a$)

$$I_z = I_{z1} + I_{z2} + I_{z3} + I_{z4} \quad I_{z1} = I_{z4} \quad I_{z2} = I_{z3}$$

$$\begin{aligned}
 I_z &= \left[\frac{1}{12} t (12a)^3 + \left(\frac{a}{2} \right)^2 (2at) \right] x_2 + \left[\frac{1}{12} at^3 + \frac{a}{6} t (at) \right] x_2 \\
 &= \frac{8}{6} a^3 t + a^3 t + \underline{a^3 t} \\
 &= \frac{17}{6} a^3 t
 \end{aligned}$$

$$I_{yz} = \cancel{2at} \left(-\frac{a}{2} \right) \left(\frac{a}{2} \right) \cancel{x_2} = -a^3 t$$

$$\tan 2\varphi^* = \frac{2I_{yz}}{I_y - I_z} = \frac{-2a^3 t}{-\frac{5}{3}a^3 t} = \frac{6}{5}$$

$$2\varphi^* = 50.19^\circ$$

$$\varphi_1^* = 25.10^\circ$$

$$\varphi_2^* = 115.10^\circ$$